

the centre of financial econometrics over the past 50 years. For the purposes of this chapter, we shall define the volatility at time t , as the conditional variance of y_t , i.e.

$$\sigma_t^2 = \text{Var}(y_t \mid \sigma_t^2),$$

where y_t denotes the return at time t . Properties of returns and volatility are the subject of many studies, sometimes referred to as *stylised facts* of the returns and volatility. The following is a short summary, and for more information the reader is referred to Tsay (2002, Chapter 1).

1. the historical mean of the returns is very close to zero;
2. the returns have heavy tails, typically heavier than the Gaussian and Student t distributions;
3. the distribution of the returns is asymmetric, typically following a left skew distribution;
4. the volatility is time-varying;
5. the volatility is observed in clusters, i.e. there are periods of time with a certain level of volatility.

Skewness in the returns suggests that positive and negative returns are not equally likely. In periods of market decline (as for example in the credit crunch of 2008), investors lose confidence and negative returns exhibit low frequency; instead, in periods of growth, investments increase and the positive returns are more likely. The characteristics of skewness are studied by many econometricians, see e.g. Bakshi et al. (2003) and the references therein. The tails of the returns are heavier than the Gaussian and the Student t distributions. Hence there is significant mass under negative and positive returns of considerable magnitude, in either side. This is demonstrated empirically in many studies and theoretically in others, see e.g. Bingham and Kiesel (2002). For a detailed discussion of the characteristics of returns and their distribution the reader is referred to Tsay (2002, Chapter 1).

7.3.2 Stochastic Volatility Model

Engle (1982) and Bollerslev (1986) introduced the generalised autoregressive conditional heteroskedastic (GARCH) models to estimate the volatility σ_t^2 . GARCH and their numerous generalisations assume that the volatility is a function of the lagged squared returns. Let y_t denote the log-returns and σ_t^2 the volatility at time t , and let $\{\epsilon_t\}$ be a sequence of independent innovations. The GARCH specification sets

$$y_t = \sigma_t \epsilon_t, \quad \epsilon_t \sim N(0, 1)$$

and

$$\sigma_t^2 = \alpha_0 + \alpha_1 y_{t-1}^2 + \cdots + \alpha_p y_{t-p}^2 + \beta_1 \sigma_{t-1}^2 + \cdots + \beta_q \sigma_{t-q}^2, \quad (7.25)$$

for some positive integers p and q and some parameters α_i and β_j with $\alpha_0 > 0$ and $\sum_{i=1}^p \alpha_i + \sum_{j=1}^q \beta_j < 1$. Given a set of observed returns $y_1, \dots, y_n$, estimates of α_i and β_j may be obtained by maximising the log-likelihood function

$$\log p(y_1, \dots, y_n \mid \alpha_i, \beta_j) = -\frac{n}{2} \log(2\pi) - \frac{1}{2} \sum_{t=1}^n \log \sigma_t^2 - \frac{1}{2} \sum_{t=1}^n \frac{y_t^2}{\sigma_t^2},$$

where σ_t^2 is given in (7.25). Since σ_t^2 is a deterministic function of α_i and β_j , we can readily obtain the maximum likelihood estimate of σ_t^2 by plugging in (7.25) the maximum likelihood estimates of α_i and β_j . In this specification the values of p, q are assumed known, but it may be possible to be estimated from the data. Other specifications of the GARCH model involve a number of improvements including replacing the Gaussian distribution of the innovations ϵ_t by a Student t distribution. For a good review of GARCH-type volatility models, the reader is referred to Tsay (2002).

As pointed out in Harvey et al. (1994), the GARCH family of volatility models suffers from three drawbacks: (1) the volatility is given as a deterministic function of past squared returns, (2) it is not parsimonious as it includes many parameters and the likelihood maximisation might suffer from local maxima and (3) it does not offer good generalisations to multivariate models, without the compromise of the curse of dimensionality (too many parameters). As a result in the 90s efforts were devoted to developing alternative models, which would overcome the above drawbacks. The idea of treating the volatility σ_t^2 as a stochastic process was advocated by a number of authors, see e.g. Harvey et al. (1994) and Jacquier et al. (1994). Both of these studies adopt the stochastic volatility model described in Sects. 1.3.3 and 6.3, but they develop different inference. In the sequel we shall discuss the basic model adopted by many authors, see e.g. Jacquier et al. (1994) and Kim et al. (1998).

Consider that log-returns $\{y_t\}$ are generated from the model

$$y_t = \exp(h_t/2)\epsilon_t, \quad \epsilon_t \sim N(0, 1), \quad (7.26a)$$

$$h_t - \mu = \phi(h_{t-1} - \mu) + \omega_t, \quad \omega_t \sim N(0, \sigma_\omega^2), \quad (7.26b)$$

where $\{\epsilon_t\}$ is an independent sequence of innovations, $\{\omega_t\}$ is an independent sequence of innovations and ϵ_t is independent of ω_s , for any t, s . The model can be cast in state space form as in Sect. 1.3.3. From the observation model (7.26a), the returns are distributed as

$$y_t \mid h_t \sim N[0, \exp(h_t)];$$